Synthetic I/O Diagnostics Guide

Page 1: Digital Input Not Changing State

If a digital input remains fixed, separate the field signal, wiring and controller interpretation before replacing components.

- Confirm the expected field device state and compare it with the electrical signal at the input terminal using approved test practices.
- Check terminal assignment, wiring continuity and the configured input channel.
- Review input diagnostics for channel faults and confirm that the software tag maps to the intended physical input.
- Record the observed field state, terminal state and software state so the discrepancy is clear.

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Page 2: Output Channel Fault

An output fault should be investigated with the load isolated according to site procedures and with attention to diagnostic indicators.

- Review the output module diagnostic status and identify the affected channel.
- Check the documented load rating and the connected load for an overload or short-circuit condition.
- Inspect wiring and terminal condition before resetting the fault.
- After correction, verify the output state under controlled conditions and record the result.

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Page 3: Unstable Analogue Sensor Signal

A fluctuating analogue signal may originate from the sensor, wiring, grounding, scaling or process conditions.

- Compare the raw input value with the expected sensor range and the configured engineering-unit scaling.
- Inspect shielding, grounding and cable routing for sources of electrical noise.
- Check for loose terminals and confirm the sensor supply is stable.
- Trend the raw value over time before and after each change so the effect can be evaluated.

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